Status	Finished
Started	Friday, 16 October 2026, 12:44 PM
Completed	Friday, 16 October 2026, 12:53 PM
Duration	9 mins 8 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Dividing the list in half repeatedly
b. Skipping every second element
c. Sorting the list before searching
d. Checking each element sequentially ✓

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
b. There must be a mechanism to access middle element directly
c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
d. Must use a sorted array

Question **3**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 94 and 99
b. 90 and 100
c. 89 and 94
d. 90 and 99 ✓

Question **4**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. At the beginning of the list
b. In the right sublist ✓
c. In the middle of the list
d. In the left sublist

Question **5**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection
b. Complexity
c. Insertion
d. Bubble ✓

Question **6**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search has a lower time complexity on large, sorted lists ✓
b. Binary search does not require dividing the list
c. Binary search works on unsorted lists
d. Binary search can find multiple instances of the target element

Question **7**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and unsorted
b. When the list is large and sorted
c. When the list is small and unsorted ✓
d. When the list is small and sorted

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is sorted ✓
b. When the list is unsorted
c. When the list is very small
d. When the list contains duplicate elements

Question **9**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The list is sorted
b. The search ends successfully ✓
c. The search continues in the left sublist
d. The search continues in the right sublist

Question **10**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the search algorithm finishes
b. When the list contains duplicate elements
c. When the element is found in the list ✓
d. When the list is sorted

Question **11**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Binary search
b. Jump search
c. Linear search ✓
d. Hashing

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It requires the list to be sorted ✓
b. It can only be applied to large lists
c. It does not work with negative numbers
d. It is slower than linear search for small lists

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present
Number values should only be present

b. The list must be sorted
c. There should be direct access to the middle element in any sublist
d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **14**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1
b. n/2
c. n ✓
d. log n

Question **15**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. n ✓
c. 1
d. log n